

CSCE 20004 – Programming Foundations I
Programming Project 3
Due 09/27/2025

1. Problem Statement:

The goal of this programming assignment is to give students experience working with different types of loops found within C++ and many other programming languages. In this assignment, students will draw various shapes using asterisks (*) and dot symbols (.)

There are two input variables to your shape drawing program:

- The shape the user wants to draw.
- The size of the shape.

Using this information, your program will use loops to produce a drawing in the terminal. For example, if shape=triangle and size=5, then the image below shows that there would be a triangle with size of 5 in the terminal.

```
Choose what to draw:
1. Triangle
2. Diamond
3. Checkerboard
4. Exit
Enter your choice (1-4): 1

Enter the size of the triangle (positive number): 5

*
* *
* * *
* * * *
* * * * *
```

2. Design

For this assignment, the first task is reading user input. You must design the prompts to ask the user for input. Then, you must design user input loops with error correction to make sure the user enters valid values. You can assume the users will always enter integers, so you do NOT need to handle bad inputs like “hello”.

Your final task is to design the loop that draws the shapes - triangle, diamond, and checkerboard. It is significantly easier to use a loop to print the '*' and/or '.' based on the user inputs. Your loop should stop printing when the shape is finished drawing.

3. Implementation

Since you are starting with a "blank piece of paper" to implement this project, it is very important to develop your code incrementally writing comments, adding code, compiling, debugging, a little bit at a time. This way, you always have a program that "does something" even if it is not complete.

As a first step, it is always a good idea to start with an empty main function, and add the code to read input data from the user, and then simply print these values back out again. Once this part is working, you can start performing calculations with the input data.

4. Testing:

Test your program with a variety of inputs to check that it operates correctly for all of the requirements listed above. Once your program is running correctly, you should run it several times with different values for shapes and sizes. Include the results of your experiments in your project report.

5. Documentation:

When you have completed your C++ program, write a short report using the "Programming Project Report Template" describing what the objectives were, what you did, and the status of the program. Does your program work properly for all test cases? Are there any known problems? Save this project report in a separate document to be submitted electronically.

6. Sample Outputs:

Diamond Example -

```

Choose what to draw:
1. Triangle
2. Diamond
3. Checkerboard
4. Exit
Enter your choice (1-4): 2

Enter the size of the diamond (positive number): 5

  *
 * *
* * *
* * * *
* * * * *
* * * * *
 * * *
  * *
   *

```

Checkerboard Example:

```

Choose what to draw:
1. Triangle
2. Diamond
3. Checkerboard
4. Exit
Enter your choice (1-4): 3

Enter the size of the checkerboard (positive number): 5

* . * . *
. * . * .
* . * . *
. * . * .
* . * . *

```

7. Project Submission:

In this class, we will be using electronic project submission to make sure that all students hand their programming projects and labs on time, and to perform automatic plagiarism analysis of all programs that are submitted.

When you have completed the tasks above, go to Blackboard to upload your documentation (a single docx or pdf file), and your C++ program (a single cpp file). Do NOT upload an executable version of your program.

The dates on your electronic submission will be used to verify that you met the due date above. All late projects will receive reduced credit:

10% off if less than 1 day late,

20% off if less than 2 days late,

30% off if less than 3 days late, no credit
if more than 3 days late.

You will receive partial credit for all programs that compile even if they do not meet all program requirements, so handing projects in on time is highly recommended.

8. Academic Honesty Statement:

Students are expected to submit their own work on all programming projects, unless group projects have been explicitly assigned. Students are NOT allowed to distribute code to each other, or copy code from another individual or website. Students ARE allowed to use any materials on the class website, or in the textbook, or ask the instructor and/or GTAs for assistance.

This course will be using highly effective program comparison software to calculate the similarity of all programs to each other, and to homework assignments from previous semesters. Please do not be tempted to plagiarize from another student.

Violations of the policies above will be reported to the Provost's office and may result in a ZERO on the programming project, an F in the class, or suspension from the university, depending on the severity of the violation and any history of prior violations.